

A particle model of hexane removal and water uptake in soybean meal desolventizing

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Abstract

In a desolventizer-toaster (DT), steam desolventization of oilseed meal couples hexane removal, water uptake and heat transfer. We formulate a conservative particle model containing free liquid hexane, binary pore gas, retained water and hexane, and a separate external water film. Component and energy jump conditions determine how the liquid-hexane interface recedes; after the core disappears, the same inventories continue to evolve in a stationary radial particle. Published thermodynamic and retention properties, together with declared transport coefficients, parameterize the model. Converged discrete calculations conserve water, hexane and energy through all three regimes. The formulation is exercised in two prescribed baths, in which the surrounding gas history is specified rather than solved. In the first, a water-bearing soybean particle in Faner’s pure-hexane vapour at 120 °C falls from 0.20 to 0.10 kg hexane/kg dry meal in 17.7 s, compared with 25.4 s measured. The source reports preheating, but not the exact initial temperature, moisture or mass-weighted particle sizes. In the second, a particle followed through a 1766.5-s prescribed DT history leaves with 18.2% wet-basis moisture, 111 ppm wet-basis hexane and a mean temperature of 105.0 °C. The particle response is benchmarked against available literature experiments and published industrial outlet ranges, without fitting the falling-rate trajectories or DT outlet values. This common formulation provides the particle-level basis for a DT digital twin that can shorten process design and support model-based control.

Keywords: desolventization; soybean meal; n-hexane; moving-front model; sorption; food process modeling

Practical Applications. A prescribed-bath experiment lets a process engineer examine solvent loss, moisture uptake and temperature without first solving a complete desolventizer-toaster (DT). The same particle equations describe a pure-hexane drying experiment and a steam-rich DT scenario. In the pure-hexane case, the calculated particle dries faster than the measured charge; better size, exposure and temperature information is needed to explain the difference. In the DT case, the particle reaches 18.2% wet-basis moisture with liquid-water imbibition included; the uptake rate rests on one same-material measurement. The model returns signed water, hexane and energy fluxes that can later be coupled to conservative tray balances.

1 Introduction

Solvent extraction leaves oilseed meal wet with hexane. In a desolventizer-toaster (DT), live steam strips the solvent while heated trays supply additional energy (Karnofsky, 1985; Sipos & Witte, 1961; Witte, 1995). Residual solvent matters for safety and emissions; condensed water must later be removed, and the meal’s thermal history affects protein quality (Kemper, 2013; Mustakas, Moulton, Baker, & Kwolek, 1981; U.S. EPA, 1995). The engineering question is how to supply enough heat and residence time for solvent removal and toasting without excessive steam use, later drying duty or damage to meal quality (Kemper, 2013; Witte, 1995). A particle model must therefore connect solvent removal, moisture uptake and heat transfer throughout the transition from externally wetted meal to a particle containing only retained material and pore gas.

These processes are coupled. Evaporation consumes energy at the particle surface or an internal liquid boundary. Incoming steam can condense, and both retained water and residual hexane contribute to storage and transport. Liquid-water imbibition also differs from vapour sorption. Experiments on soy protein isolates show that imbibition combines sorption, capillarity and interstitial retention, whereas vapour sorption describes a different hydration experiment (Jovanovich, Puppo, Giner, & Añón, 2003). Those measurements motivate treating liquid contact separately from vapour exposure; their capacities and rates cannot be transferred directly to hot, hexane-bearing extracted meal.

Published desolventization models emphasize different parts of this sequence. The superheated-hexane experiments and receding-front model of Faner, Perez, and Crapiste (2019) describe a heat-fed liquid-hexane core. The transport account of Cardarelli, Crapiste, and Mattea (2002) resolves a diffusion-controlled drying tail. Coletto, Bandoni, and Blanco (2022) couples a transient radial particle to a steady axial bed in the diffusion-controlled zone, treating the preceding flashing zone separately. Here we combine component and energy conservation at a receding liquid-hexane interface with binary pore gas, matrix retention and a separately conserved external water film. After the core disappears, the same stored components and energy continue to evolve in a stationary radial particle.

In the first of two prescribed baths, a water-bearing particle in pure superheated hexane is compared with the soybean experiment of Faner et al. (2019); pure supplied vapour does not imply a water-free meal. In the second, a particle is followed from feed through a prescribed steam-rich DT history. Prescribing the bath isolates the particle response, so the surrounding vapour composition and temperature need not be predicted. In a future coupled tray calculation, the signed water, hexane and energy fluxes would enter the gas and tray balances, which would determine the next local bath state.

The two comparisons rest on different kinds of evidence. In the Faner experiment, the measured constant-rate duty informs external heat supply; later loading and temperature readings provide the comparisons. Faner reports preheating in the apparatus and particle-size summaries, but not the exact initial temperature, initial meal moisture or a mass-weighted size histogram. The DT history is an engineering scenario; published outlet ranges provide process context rather than measurements of the same particle and operating point. Liquid-water imbibition is part of the particle formulation, and its transport coefficients are declared rather than measured. Numerical verification and complete time integration make these assumptions inspectable; independent measurements are needed to establish their predictive accuracy.

The paper has two goals: to state one conservative particle formulation and to exercise it in both baths. Section 2 defines the particle equations and phase regimes, and Section 3 their numerical integration, admissibility checks and verification. Section 4 presents the two experiments, Section 5 discusses their meaning for DT practice and the measurements needed next, and Section 6 concludes.

The Supplementary Material contains detailed source readings, parameters, numerical evidence and the separate scalar Fickian comparison with the earlier Faner measurements.

2 Particle model

This section states the particle model that the rest of the paper exercises: the phase regimes of one particle, the balances in each region, the conditions that close the moving front and the properties that enter them. Readers mainly interested in the process results can read Section 2.1 and then continue with Section 4.

2.1 Particle geometry, inventories and phase regimes

The model follows one particle through three regimes, defined by where liquid hexane remains. Hexane is stored as external free hexane, hexane in a liquid-hexane core, retained hexane and pore gas. External free hexane wets the surface in regime A. Its depletion allows the liquid-hexane core to recede in regime B, leaving a dry shell. Throughout this paper, *dry shell* means depleted of free liquid hexane, not free of water or retained hexane. Regime C begins when the liquid-hexane core disappears. Steam can form an external water film, from which *liquid-water imbibition* transfers water into the particle’s retained-water inventory. Retained water includes effective matrix and interstitial retention; it does not imply that all water is molecularly bound.

We approximate the particle by a porous sphere of fixed radius R_p . This geometry resolves radial heat and mass transport with one length scale, as in the particle descriptions of Faner et al. (2019) and Cardarelli et al. (2002). Actual meal particles are irregular; the radius selected for each experiment is therefore stated with its source and interpretation. Fixed radius and porosity separate transport from swelling and shrinkage, for which no constitutive laws are supplied here. The particle’s total n-hexane loading X_h (every loading here in kg per kg of dry, water- and solvent-free meal) is compared with two thermodynamic reference loadings at the particle’s temperature. The first is the critical loading $X_c(T)$, at which mobile liquid stops reaching the outer surface; it is set on the transition loading Faner et al. (2019) measured (Table 1). The second is the equilibrium loading $X_e^*(T, P)$, at which no condensed hexane remains. The star denotes evaluation at the pressure-feasible saturated activity a^* of Eq. (1). In regime A, while external free hexane remains, bulk liquid wets the outer surface at unit hexane activity and the resistance is external. In regime B ($0 < s < R_p$), liquid is present only behind a receding interface, and removal is limited by the heat reaching that interface through a growing dry shell. In regime C, no liquid-hexane core remains: $s = 0$. Retained material and pore gas continue to obey the same local component and energy balances. Core extinction is a geometric conservation event; a particle-mean loading is not used to prescribe it. The saturated equilibrium loading is a thermodynamic reference rather than a uniform-particle transition rule (Figure 1).

Figure 1 follows external free-hexane depletion ($X_f = 0$) and subsequent core disappearance ($s = 0$). Purple represents hexane and blue represents water; orange arrows show incoming heat. The dashed blue band is the optional external water film, through which hexane can escape in the proposed closure. Liquid-water imbibition operates in the dry shell connected to that film, with no additional liquid-water flux into the liquid-hexane core. Dots and diamonds represent retained water and retained hexane, respectively. Geometry and film thickness are schematic. Fluxes are outward-positive: j is mass-based and N molar, with $h_k^{\text{mass}} = \bar{h}_k/M_k$; subscripts p and g identify the particle and gas sides of the surface.

The particle model is written so that it can later be coupled to a tray model. For prescribed gas and heat-transfer conditions, the particle balances return outward-positive surface fluxes $j_w(R_p)$,

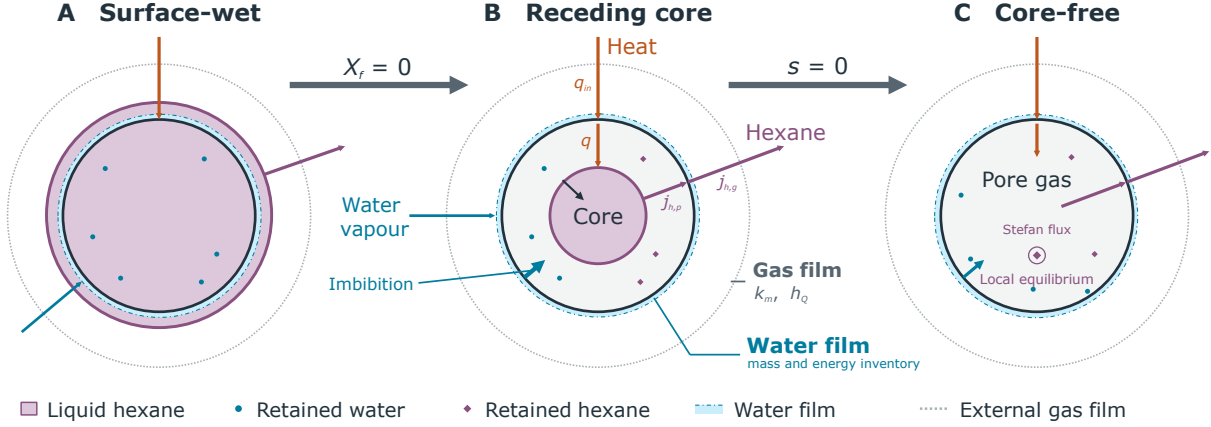


Figure 1: Particle regimes and transport pathways.

$j_h(R_p)$ and $J_E(R_p)$ on one caloric datum. These are responses to the evolving interior, rather than imposed drying rates. For particle classes c with total exposed areas $A_{ex,c}$, the rates delivered to a future tray gas balance would be $\sum_c A_{ex,c} j_{k,c}(R_{p,c})$ for component k and $\sum_c A_{ex,c} J_{E,c}(R_{p,c})$ for total energy. Condensation has a negative water rate; each gas source has an equal and opposite entry in the particle balance. The tray would return its updated gas state to the particles. In this paper, the gas and bed conditions along the equipment trays are prescribed instead. This separates the particle response from countercurrent gas mixing and tray residence, which is useful for testing the particle equations before coupling them to DT design calculations; steam consumption itself requires those equipment balances (Kemper, 2013).

The equilibrium boundary is evaluated at the *pressure-feasible* hexane activity

$$a^*(T, P) = \min\left(1, P/P_{sat}^h(T)\right), \quad (1)$$

with P the total pressure and P_{sat}^h the n-hexane vapour pressure, both in pascals. Equation (1) is an ideal-mixture diagnostic. The pore and interface equilibrium used in the calculations uses binary-virial gas fugacities and the pure-liquid equation-of-state fugacities on the same pressure and caloric datum. It does not omit those corrections when locating phase boundaries. Above the solvent's boiling point at the bed pressure, an equilibrium loading taken at unit activity is overstated. Evaluated with (1), the two boundaries stay one to two orders of magnitude apart and separate further as the temperature rises; the paper uses only that *ordering* (Sec. S6).

The A–B transition occurs when the external free-hexane inventory X_f is exhausted. While $X_f > 0$, liquid wets the surface and the front stays at $s = R_p$; once $X_f = 0$, it may recede and leave a dry shell. This sequential-drainage condition is stated in Eq. (15). The reference critical loading of 0.20 kg/kg measured by Faner et al. (2019) sets the effective critical-volume capacity at the reference liquid density, 615 kg/m³. At the particle's actual temperature, $X_c(T) = \kappa_v \varepsilon_g \rho_{h,l}(T, P) / \rho_{dm,p}$, with $\kappa_v = 2.675$ fixed by that reference capacity. The liquid density comes from the same equation of state used in the component and energy balances. The depletion loading therefore has a prescribed temperature dependence, while the time at which it is reached follows from the regime-A inventory balance under the prescribed gas and heat history. The component and energy jumps then determine the recession speed and the dry-side hexane flux at the front. The shell balances and surface conditions determine the corresponding particle-surface exchanges. In the discrete model the front first departs over an accepted time step, giving a finite shell thickness without assigning the shell the width of one radial cell (Sec. S4). The A–B boundary is thus a surface-state constraint, with no fitted

time or drying-rate selector. The stationary binary regime-C closure starts after the liquid-hexane core has vanished. In a calculation, the crossing *loading* therefore reproduces a supplied value; the crossing *time* is the tested prediction, conditional on the external history (Section 4.1).

Heat must first cross the gas-side resistance and then the dry shell to reach the receding front. In the reported scale comparison, the external thermal resistance dominates until the core radius falls to about $0.28R_p$; the dry-shell resistance dominates later (Sec. S1). The present total-vapour closure adds no pressure-driven transport resistance, so front recession is heat-limited. This distinction matters when interpreting falling-rate data: the 2019 Faner records support a heat-limited description, whereas the older Faner records and the mechanism of Cardarelli et al. (2002) support a diffusion-limited description. The separate scalar Fickian comparison in Eq. (S2) represents the latter; it does not replace the shared particle law in either experiment (Sec. S3).

Temperature and composition also vary over different radial scales. The dry-shell Lewis number of 223.25 requires their profiles to be reconstructed separately near the moving front (Crank, 1956; Perré, Rémond, Almeida, Augusto, & Turner, 2023).

2.2 Local balances and region closures

Inside each region the model is a set of conservation laws with no source terms, closed differently in the liquid-hexane core and in the dry shell. The sphere holds two conserved mobile components, water and n-hexane, and an immobile composite dry meal of local density $\rho_{\text{dm},p}$ with a residual-oil mass fraction w_o . A sharp interface at radius $r = s(t)$, with t the time, separates a liquid-hexane core (w) from a dry shell (d); the region label α takes either value. Radial fluxes are positive outward, and recession means $\dot{s} < 0$. The total radial energy flux is defined once, on both sides, as $J_E^\alpha = q^\alpha + \sum_k \bar{h}_k^\alpha j_k^\alpha$, with q the conductive heat flux, j_k the radial flux of component k and $\bar{h}_k$ the partial molar enthalpy on the common datum of Section 2.7. In each smooth region

$$\frac{\partial C_k^\alpha}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 j_k^\alpha) = 0, \quad k \in \{\text{w}, \text{h}\}, \quad (2)$$

$$\frac{\partial u^\alpha}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 J_E^\alpha) = 0, \quad j_{\text{dm}} = 0, \quad j_{L_o} = 0, \quad (3)$$

with C_k the total local concentration of component k over all of its phases, u the total local internal energy density, and j_{dm} and j_{L_o} the fluxes of the immobile dry meal and residual oil. Retention, evaporation and condensation redistribute a component between local phases and are therefore *not* sources in (2): no phase-change term appears in the smooth-region equations (Sec. S1).

Two statements close the liquid-hexane core. First, no relative n-hexane flux is carried inside the mobile-liquid region: at activation, every material shell stores the historical label $X_{\text{h,w}}^a = X_c(\theta_a)$, the critical loading at its temperature at activation θ_a , and thereafter

$$\frac{D_s}{Dt} (\rho_{\text{dm},p} X_{\text{h,w}}^a) = 0 \quad (4)$$

following the solid. Thus $X_{\text{h,w}}^a$ is the apparent mobile-liquid hexane loading of the liquid-hexane core, stored at activation and distinct both from the particle's total loading X_h and from the retained hexane of the dry shell. The core is liquid-filled, the hexane activity is unity, and the non-negativity of the mobile liquid is an admissibility condition, not a clipped quantity. Second, the core is resolved, not lumped, with specific energy per kilogram of dry meal

$$e_w = h_{\text{dm}}(T, w_o) + X_w h_{\text{w,l}}(T, P) + X_{\text{h,w}}^a u_{\text{h,l}}(T, P) - B_h(a_h=1, T, w_o), \quad (5)$$

with h_{dm} the specific enthalpy of the dry meal, X_w the total water loading, and $h_{\text{w,l}}$ and $u_{\text{h,l}}$ the specific enthalpy of liquid water and the specific internal energy of liquid n-hexane, both in J/kg.

Retained and mobile hexane are not carried separately in this energy, so the binding potential B_h is counted once and the repartition between them is source-free.

The dry shell stores each component in two forms, pore gas and a retained contribution, with distinct storage terms and distinct thermodynamics; no retained-only symbol includes pore gas. Transport consists of one thermodynamic-force pore flux, one retained-matrix water flux and one algebraic total flux,

$$J_w = -J_h = -L_{wh}^{\text{eff}} \nabla_{\text{comp}} \left[\frac{\mu_w - \mu_h}{RT} \right], \quad (6)$$

$$N_{w,\text{ret}} = -L_{w,\text{ret}}^{\text{eff}} \nabla_{\text{comp}} \left[\frac{\mu_{w,\text{ret}}}{RT} \right], \quad N_i^g = y_i N_t + J_i, \quad (7)$$

with μ_w and μ_h the pore-gas chemical potentials of water and n-hexane and $\mu_{w,\text{ret}}$ that of retained water (J/mol), R the gas constant, L_{wh}^{eff} and $L_{w,\text{ret}}^{\text{eff}}$ the pore and retained-matrix mobilities (mol/(m s)), y_i the pore-gas mole fraction of component i , and the fluxes J_i , $N_{w,\text{ret}}$, N_i^g and N_t molar (mol/(m² s)). The total (Stefan) flux N_t is recovered algebraically from the summed continuity equations rather than fitted. It rests on an Onsager mobility whose degeneracy vanishes exactly at either purity and needs no composition floor (Krishna & Wesselingh, 1997). The shell energy flux carries each stream exactly once, with no latent, binding, pressure-volume or Stefan-energy correction on top (Sec. S1.1). We retain composition-driven mass transport and conductive heat transport, and neglect independent Soret terms because no thermal-diffusion coefficients for this meal are supplied. This is a constitutive approximation; it does not show that thermal diffusion is negligible. Here ∇_{comp} denotes the composition part of the fixed- (T, P) chemical force, not the full gradient of a temperature-dependent standard chemical potential. At ordinary faces, the isothermal composition secant is evaluated at a common face temperature. The moving-interface reconstruction integrates that composition derivative along the actual linear temperature/composition path,

$$G_{\text{comp}}(\Xi) = \frac{1}{d} \int_0^1 \frac{\partial \Xi}{\partial y_h} \Big|_{T(\lambda), P} \frac{dy_h}{d\lambda} d\lambda, \quad \Xi = \ln(f_w/f_h) \text{ or } \ln(f_w/f_{\text{ref}}). \quad (8)$$

Here d is the face separation and f_i is the equation-of-state fugacity. Energy transport remains non-isothermal and includes component enthalpies. This is the declared nominal transport approximation in both prescribed-bath experiments. The pressure variation omitted by taking N_t from continuity is estimated from Darcy's law across the dry shell,

$$\Delta P \approx \frac{N_t RT}{P} \frac{\mu_g}{B} (R_p - s), \quad (9)$$

with μ_g the pore-gas viscosity and B the Darcy permeability. For an illustrative vapour-flux bound, take $|N_t| = 0.5 \text{ mol}/(\text{m}^2 \text{ s})$; this is a supplied sensitivity scale, not the measured or calculated mean flux of the new DT history. With this flux bound, $\mu_g = 10^{-5} \text{ Pa s}$, T at the heteroazeotrope and a shell 1 mm thick, the estimate gives ΔP of about 0.15 kPa, 1.5 kPa and 15 kPa at $B = 10^{-12}$, 10^{-13} and 10^{-14} m^2 . A capillary bundle $B = \varepsilon_g r_{\text{pore}}^2 / (8\tau)$ at the particle porosity 0.14 and a tortuosity of three is about 10^{-13} m^2 for a 5 μm pore (ΔP about 1 kPa on the same bound) and about $6 \times 10^{-15} \text{ m}^2$ for a 1 μm pore (ΔP about 25 kPa). Below about 10^{-15} m^2 the estimate exceeds the bed pressure. The isobaric closure therefore assumes that the vapour crosses a connected pore space of several micrometers. It does not apply to a path confined to sub-micrometer pores, and no measured permeability of this meal is supplied here. These fluxes do not reduce to a scalar Fick law for n-hexane. Summing (7) over the two components gives $N_w^g + N_h^g = N_t$, fixed by continuity and not by a gradient. With the mobility $L_{wh}^{\text{eff}} = c y_w y_h D_b$, where c is the molar density of the pore

gas and D_b the binary pore coefficient, (6) becomes a binary Fick law with coefficient D_b only at zero total flux and constant total concentration. In a pure n-hexane pore gas the binary flux vanishes, and the n-hexane flux is N_t itself (Secs. S1.1 and S1.1). The scalar Fick description used for the older drying records is a separate comparison with an effective transport coefficient, given in Eq. (S2). In both prescribed baths, regime C instead integrates the stationary radial binary component and total-energy balances. Its initial inventories and temperatures come from the conservative core-extinction projection. The same particle equations therefore carry solvent removal through core disappearance and the remaining drying tail.

The value of the coupled formulation lies in its accounting; it does not claim that one coefficient represents every possible falling-rate mechanism. Water and n-hexane share the pore-gas and energy balances, while water in the retained matrix has its own transport path and retained hexane contributes to both storage and binding energy. A change in the rate of evaporating hexane therefore also changes the heat carried out, the temperature and the conditions under which water can condense or desorb; none can be added as an independent source after solving a hexane-only particle. The immobile wet-core liquid and the sorbate in the dry shell are distinct inventories. While the front is still moving, recession is limited by heat. After the front vanishes, regime C carries the remaining hexane in the same stationary binary radial model. Equation (S2) is the radial problem used for the comparison with the slower literature curves. The comparison does not establish that the heat-limited front and the diffusion-limited tail are asymptotic branches of one fully identified rate law.

2.3 Front conditions

The front conditions fix the front velocity from the heat that reaches the front. For a spherical control volume with moving inner and outer radii $r_1(t)$ and $r_2(t)$, conservation of either component and of total energy requires

$$\frac{d}{dt} \left(4\pi \int_{r_1}^{r_2} C_k r^2 dr \right) = -4\pi r_2^2 (j_k - \dot{r}_2 C_k)_{r_2} + 4\pi r_1^2 (j_k - \dot{r}_1 C_k)_{r_1}, \quad k \in \{w, h\}, \quad (10)$$

$$\frac{d}{dt} \left(4\pi \int_{r_1}^{r_2} u r^2 dr \right) = -4\pi r_2^2 (J_E - \dot{r}_2 u)_{r_2} + 4\pi r_1^2 (J_E - \dot{r}_1 u)_{r_1}. \quad (11)$$

The terms proportional to $\dot{r}_1$ and $\dot{r}_2$ account for material swept through the moving boundaries. Shrinking this volume onto $r = s(t)$ gives the Rankine–Hugoniot conditions for both components and total energy (Crank, 1984),

$$\llbracket j_k - \dot{s} C_k \rrbracket_w^d = 0, \quad k \in \{w, h\}, \quad (12)$$

$$\llbracket q + \sum_k \bar{h}_k j_k - \dot{s} u \rrbracket_w^d = 0, \quad (13)$$

with $\llbracket \cdot \rrbracket_w^d$ the dry-side value minus the wet-side value. Two different wet-side states occur in these conditions and must be kept apart. The first is the equilibrium interface trace, which the interface conditions solve for. The second is the *donor* state, the one-sided limit of the wet-core fields themselves. Every term multiplying $\dot{s}$ is a stored state, swept at the state of its own side. With the zero wet-side relative hexane flux of (4), the hexane jump fixes the dry-side flux that the moving front produces, $j_{h,d} = \dot{s}(C_{h,d}^\Gamma - C_{h,w}^b)$. The wet-side donor value, the historical label stored at activation rather than a re-solved critical loading, must exceed the dry-side trace. Substituting this flux into (13) moves the term $\bar{h}_{h,d} j_{h,d}$, which is proportional to $\dot{s}$, into the denominator. The

front velocity is then

$$\dot{s} = - \frac{[[q]]_w^d + \bar{h}_{w,d} j_{w,d} - \bar{h}_{w,w} j_{w,w}}{\bar{h}_{h,d} (C_{h,d}^\Gamma - C_{h,w}^b) - (u_d^\Gamma - u_w^b)}, \quad (14)$$

a heat-supply to stored-state-jump quotient in its donor-restricted form. The numerator is the conduction jump, dry side minus wet side, plus the enthalpy carried by the water flux. That water flux is evaluated from the water jump and the shell. The denominator is the energy required to sweep a volume of liquid-hexane core into the dry-side state. On the common datum of Section 2.7 it contains the latent heat of the hexane removed, a smaller ideal-gas correction and the binding energy already stored in u (Sec. S1). The superscript Γ marks the equilibrium interface trace, b the donor state, and $\bar{h}_{h,d}$ the partial enthalpy of n-hexane on the dry side. This is where a heat limit and a transport limit could meet in one expression. In this formulation they do not. The dry-side flux is not throttled by (6), which carries only the separation of the two vapours, and the total flux is algebraic. The numerator, the heat reaching the front through the film and the dry shell in series, therefore sets the front velocity wherever the vapour is swept outward faster than the two vapours separate. In a march inside the band, the shell Péclet numbers are of order 10^3 (Sec. S6). A transport-controlled recession needs a finite resistance on the total flux. Equation (S2) would supply one, but this formulation does not apply it at the moving front: the front velocity contains no Darcy or Knudsen term. Regime C begins only after that front has vanished and retains the binary radial closure. The model contains no rate selector, no minimum of a diffusion rate and a heat rate, and no empirical evaporation-rate law.

2.4 Admissible states and the removal of swept material

The front conditions hold in the discrete model only if states outside the admissible set are rejected and swept material is removed at one consistent state. The interface is closed by continuity of temperature, local phase equilibrium at unit hexane activity on the wet side, and equal liquid and gas pressure; the last condition means that macroscopic capillarity is omitted. Admissibility is a complementarity system rather than a clipping rule,

$$X_f \geq 0, \quad R_p - s \geq 0, \quad X_f (R_p - s) = 0, \quad \dot{s} \leq 0, \quad (15)$$

whose first three conditions express strict sequential drainage (the external free-hexane loading X_f must be exhausted before the front may leave the outer surface) and whose fourth restricts the formulation to primary drainage. A trajectory that violates any of them, inverts the interfacial concentration jump or loses interior mobile liquid is rejected at that step. The violated bound is reported and the last accepted state is left unchanged: no second interface is created and no state is repaired to keep the integration running.

One consistency requirement needs a separate statement, because a discretization can violate it silently. When the interface sweeps material, the concentration and the energy density used to debit the swept volume must be read from the same material state. Using the equilibrium trace for one and the consumed material's bulk state for the other would imply a pre-drainage that the intraparticle Péclet number of this system excludes. Both are therefore taken at the bulk, old-time state of the consumed piece. In that state the retained water is *continuous* across the front, because hexane drains and water does not (Sec. S1.1). The formal statement is a gauge covariance: shifting the arbitrary caloric zero of water by a constant c_w must move the discrete energy residual of the interface rows by exactly c_w times their discrete water residual,

$$\frac{\partial R_E}{\partial c_w} = R_w, \quad (16)$$

identically in the discrete state. Section 3.4 uses this identity as the acceptance test for the interface treatment. The same test is available to any moving-boundary phase-change discretization: if the energy residual does not move exactly with the component residual when that component's caloric zero is shifted, latent or binding heat is being counted twice, or the swept material is being debited from the wrong state.

2.5 The conserved external water film

When the bath supplies condensing water, the model stores it in an external water film, an inventory separate from retained water. Steam condensation supplies heat for desolventization while increasing the water that may require removal in the downstream dryer (Kemper, 2013; Witte, 1995). Keeping the film separate lets the balances determine how much water enters the particle and how much evaporates back to the bath. We use one film temperature and inventory at the surface, rather than resolving its liquid flow or spatial thickness. Let L_w be its mass per unit particle area, $j_{k,p}$ the outward pore-side component flux, and $j_{k,g}$ the outward gas-film flux. To isolate the effect of water uptake, the external water film is assumed to add no resistance to hexane escape. This limiting case is not a measured water-film permeability; a blocking effect would require an additional constitutive law. The surface balances are

$$\frac{dL_w}{dt} = j_{w,p} - j_{w,g}, \quad (17)$$

$$0 = j_{h,p} - j_{h,g}, \quad (18)$$

$$\frac{d[L_w u_{w,l}(T_s)]}{dt} = J_{E,p} - J_{E,g}. \quad (19)$$

The gas-film closure uses an ideal binary mass-coordinate partition, with equation-of-state phase traces and partial enthalpies. For total outward mass flux j_t , conductance $K_f = \rho_f k_m$, and hexane mass fractions $Y_{h,s}$ and $Y_{h,b}$,

$$j_{h,g} = Y_{h,s} j_t + K_f (Y_{h,s} - Y_{h,b}) \Phi(j_t / K_f), \quad j_{w,g} = j_t - j_{h,g}, \quad (20)$$

$$\Phi(x) = \frac{x}{e^x - 1}, \quad \Phi(0) = 1, \quad \beta = \frac{j_{w,g} c_{p,w}^f + j_{h,g} c_{p,h}^f}{h_Q}, \quad (21)$$

$$q_{in} = h_Q \Phi(\beta) (T_b - T_s), \quad J_{E,g} = -q_{in} + \sum_k \frac{j_{k,g}}{M_k} \bar{h}_k(T_s). \quad (22)$$

The component heat capacities are component partial-enthalpy secants between surface and bulk temperatures. Both component capacity fluxes enter β , including counter-diffusion. The same analytic Ackermann expression is used within a declared band $|\beta| \leq 3$. Its largest accepted value in the recorded calculations is 1.2063, above the previously assessed film band $|\beta| \leq 1$. This wider band is a declared assumption about the film law, separate from the numerical root tolerances; it has not been demonstrated for this gas film. All advected partial enthalpies enter the two total-energy fluxes once. Positive external liquid selects unit surface water activity; with zero liquid, gas-side composition is solved from the same three balances and may remain undersaturated. Appearance and depletion are conservative events, not resets of the film mass or temperature. The interior retention capacity is a separate active set. The numerical unit-activity trace of the hexane-transparent film and its sensitivity are stated with the experiments. For the regime-A surface in the pure-hexane bath, the hexane-only surface active set sets the outward water flux to zero while external free hexane remains. The water in the particle stays stored and becomes exposed through the binary pore gas once the external free hexane is depleted.

2.6 Liquid-water imbibition from the external water film

Liquid-water imbibition is assumed to occur when the external water film contacts the dry shell. Steam condenses on the particle faster than the water can move inward, so a liquid film accumulates on the surface and is drawn into the pore structure afterwards, where the matrix and the interstitial space retain it (Karnofsky, 1985). Liquid contact also admits water that vapour sorption does not: the imbibition capacity of soy protein isolates exceeds their vapour-sorption capacity several-fold (Jovanovich et al., 2003). We represent this path as an effective finite-conductance pathway into the existing capacity-bounded retained-water inventory. The closure does not identify every absorbed molecule as strongly bound and adds no independently resolved pore-liquid domain. Vapour transport and retention remain active through the original component equations.

Let $\psi = \ln a_{w,\text{ret}}$ be the activity potential evaluated from the actual retained-water state. For the additional liquid-water mobility $\mathcal{L}_\ell > 0$, liquid contact conductance $K_\ell > 0$ and distance $d > 0$ between the outer cell and surface trace, the external contribution is

$$G_\ell = \left(\frac{d}{\mathcal{L}_\ell} + \frac{1}{K_\ell} \right)^{-1}, \quad N_\ell = -G_\ell(\psi_s - \psi_p). \quad (23)$$

N_ℓ is outward-positive in mol/(m² s). Thus $0 < G_\ell \leq K_\ell$ and $G_\ell \rightarrow K_\ell$ as $d \rightarrow 0$: when the dry shell first forms, the surface conductance stays finite. Interior faces in the dry shell use $G_\ell = \mathcal{L}_\ell/d$. This added path is absent at the remaining wet-core interface and when external liquid is absent. Surface and cell activities follow their capacity-bounded retained states; the added force is not obtained by substituting unit pure-liquid activity for the retained surface trace.

For available film stock $L_{w,\text{old}}$, inward addition is limited by

$$j_{\ell,\text{in}} \leq \max \left(0, \frac{L_{w,\text{old}}}{\Delta t} + j_{w,\text{p}}^{\text{base}} - j_{w,\text{g}} \right). \quad (24)$$

The final flux enters $j_{w,\text{p}}$ in Eq. (19) and the particle balance with opposite signs. Its donor retained-water partial enthalpy enters the energy flux once. Under the negligible-Soret approximation for this additional path, its composition-force dissipation is proportional to $RG_\ell(\psi_s - \psi_p)^2 \geq 0$. Liquid-water imbibition is an effective transport hypothesis, not a resolved capillary-pressure, hydraulic saturation or swelling model. Water displacement of pore hexane is neglected, and the film adds no resistance to hexane escape (Section 2.5).

Two coefficients set the rate: the liquid contact conductance K_ℓ at the wetted surface and the liquid-water mobility $\mathcal{L}_\ell$ inside the shell. They act in series, so an uptake measurement on one particle size fixes only their combination. We therefore set the mobility first and derive the conductance from the one same-material measurement that exists. The mobility is 1000 times the reference retained-water mobility $L_{w,\text{ret}}^{\text{eff}}$. In porous hygroscopic solids the liquid transport coefficient rises by about three orders of magnitude between the dry state and water-filled pores (Künzel, 1995); we apply the water-filled end of that range because the path is active only while a liquid film wets the shell. Above a factor of about 300 the calculation no longer depends on this choice, because the surface then controls the uptake. A tenfold increase from 1000 to 10 000 moves the DT outlet moisture by 0.07 percentage points; a tenfold decrease moves it by 0.58 (Sec. S12). The conductance is then set from the only kinetic measurement of solvent-extracted soybean meal in liquid water: 1.4-mm particles at 80 °C are saturated within the first 90-s sample (Hemmingsen et al., 2008). Reproducing that time with Eq. (23) and the retention law of Eq. (27), and scaling to 100 °C by the viscosity of water, gives $K_\ell = 0.32$ mol/(m² s) (construction in Sec. S12). With the adopted mobility, the resistance length $\mathcal{L}_\ell/K_\ell$ is 2.6 mm, 1.8 times the DT particle radius. Surface entry therefore carries about two thirds of the series resistance at that radius, and a contact-only sphere at

the DT radius fills from the feed loading to within 0.01 kg/kg of capacity in about 150 s. The value rests on declared transfer assumptions (a sphere of the sieve size, interior saturation at the first sample, viscosity scaling) and is not a measurement on DT meal. Neither coefficient was adjusted to the DT outlet, which is governed by the condensate supply and the retention capacity (Sec. S12).

2.7 One caloric datum per component

One numerical zero per component makes every latent and binding heat appear once and gives the energy balance an acceptance test. Every state function of a component is evaluated on *one* numerical zero shared by all of its phases: mobile core liquid, external free hexane, the retained phase, pore gas and bulk vapour alike. Two consequences follow. First, latent and binding effects occur exactly once, through differences of stored state plus the component enthalpy fluxes the energy flux already carries. A separate vaporization-enthalpy source, a fitted sorption-heat source or a power-law excess sorption enthalpy is therefore excluded, not merely unnecessary, because it would count the same energy twice. Second, because the zero is arbitrary, it must cancel: shifting it may change no temperature, flux, reported quantity or external energy residual. This property becomes the acceptance test of Section 3.4. For hexane, the caloric consequences of retention follow from its temperature-dependent isotherm: the integrated sorption potential, the net heat of sorption and the binding potential B_h of (5) form a Gibbs–Helmholtz pair whose two routes must agree (Sec. S1.1). The nominal retained-water energy uses liquid-water enthalpy on the same datum. Its temperature-independent Luikov activity law contributes no separate excess water-binding heat. Condensation and evaporation energy are already present through phase-state differences and component fluxes.

2.8 Sorption isotherms, property equations and parameters

The sorption isotherms and property equations close the model, and the ranges over which they were measured set where its results can be trusted. Retained n-hexane on reference soybean meal follows the Guggenheim–Anderson–de Boer isotherm fitted by Cardarelli and Crapiste (1996),

$$W_{\text{native}}(a_h, T) = \frac{X_m C x}{(1-x)(1-x+Cx)}, \quad x = K(T) a_h, \quad (25)$$

with X_m the monolayer loading and C and K the dimensionless GAB constants. Both C and K have an Arrhenius temperature dependence, and no activity clamp is applied. The sorbing sample carried reference oil, so (25) is a *total-meal* isotherm, with no duplicate oil inventory at the reference oil fraction $w_{o,\text{ref}} = 0.0195$. For an off-anchor oil fraction the existing reference-anchored composite closure is

$$W_h = \alpha_o W_{h,\text{native}} + \beta_o q_o(a_h), \quad \alpha_o = \frac{1-w_o}{1-w_{o,\text{ref}}}, \quad \beta_o = \frac{w_o - w_{o,\text{ref}}}{1-w_{o,\text{ref}}}, \quad q_o = 0.9635 a_h^{2.7036}. \quad (26)$$

The same weights give dry-composite heat capacity $\alpha_o c_{p,\text{native}} + \beta_o c_{p,\text{oil}}$. Thus β_o vanishes at the reference anchor and is a signed correction below it, as in the declared DT oil fraction. It is not an extra oil arm added to the reference total-meal isotherm (Sec. S1.2). Retained water follows the two-parameter modified Luikov form fitted to desorption data for desolventizer-outlet meal pooled with the low-temperature soybean-meal data of Pixton and Warburton (1975) (Luikov, 1975; Luz, Sousa, Jorge, & Paraíso, 2006),

$$W_w(a_w) = \frac{A_1}{1 + A_2 \ln(1/a_w)}, \quad \ln a_w = \frac{1 - A_1/W_w}{A_2}, \quad (27)$$

with a_w the water activity and A_1 (kg/kg) and A_2 (dimensionless) its two fitted parameters. The isotherm is temperature-independent in the nominal closure; the temperature-dependent form is carried as a mandatory replacement-only caloric sensitivity case (Sec. S1.2). The shared prescribed-bath case continues this same analytic expression below the evidence anchor $W_w = 0.023$, for strictly positive water, retaining $A_1 = 0.88$, $A_2 = 12.184$ and the original upper capacity 0.236 kg/kg. This is a declared constitutive extrapolation, not new retention data. The analytic log activity is used directly when exponentiating it would underflow; water is neither floored nor held constant. Exactly absent water requires a separate component limit. Neither isotherm carries a cross-coupling term. The only direct co-sorption measurements support this at low water loading, but not beyond it (Cardarelli, 1998). Both isotherms are used outside their measured domains; these uses are marked as extrapolations in Section 5 rather than smoothed (closure detail in Sec. S6). Pure n-hexane follows the technical Helmholtz equation of state of Span and Wagner (2003a, 2003b), water the IAPWS-95 reference formulation (International Association for the Properties of Water and Steam, 2018; Wagner & Pr  , 2002), and the binary gas a second-virial mixture whose cross term carries a qualification interval rather than a value (Sec. S1.2). Table 1 gives the adopted material and transport inputs with their provenance and scope. Published sorption coefficients retain their reported precision; derived quantities and declared transport inputs are rounded for presentation, while the calculations retain their full precision. Literature fits provide the thermodynamic properties; the remaining transport inputs define one common case without being adjusted to the falling-rate or DT outlet results. The transport inputs have a different evidence basis from the sorption fits. Cardarelli et al. (2002) uses a scalar particle diffusivity of 4.0×10^{-10} m²/s, based on rapeseed evidence. The adopted binary pore value, 6.0×10^{-10} m²/s, is the upper point of an engineering sensitivity range of one-half to one-and-a-half times that source value; the source supports its scale, rather than identifying binary transport in wet soybean meal. Touffet, D  leris, and Bokhove (2026) reports apparent moisture diffusivities reaching about 6.2×10^{-10} m²/s in pelleted feed at 60   C. We use that value as a reference response and derive the retained-water mobility from the coupled component balances at one reference state (Sec. S12). This avoids assigning the same apparent diffusivity independently to both water pathways. The material and temperature differences remain assumptions; neither coefficient is a measurement of soybean transport under DT conditions.

The scale of the reference mobility is consistent with the two measurements that exist on oilseed meal: vapour-sorption diffusivities of sunflower meal at 50–95   C, 10^{-12} to 1.15×10^{-10} m²/s (Cardarelli, 1998), and the soaking diffusivity of defatted soybean meal at 20–60   C, 1.1 to 2.1×10^{-11} m²/s (Mukherjee, Chakraborty, & Dutta, 2019); the loading dependence of Eq. (27) spans that range between $W = 0.05$ and 0.20. The liquid-water mobility and the liquid contact conductance follow the chain of Section 2.6: a literature limit for the mobility factor and a same-material uptake time for the conductance. The DT moisture result is conditional on these inputs, but none was adjusted to its outlet benchmark; a fourfold change in K_ℓ moves the outlet moisture by 1.3 percentage points (Sec. S12). The separate older effective-diffusivity comparator is retained only in the supplement. The bed film follows the stated correlation convention of Faner (2008) as transcribed by Coletto et al. (2022) (Sec. S1.2). The ranges in Table 1 describe source evidence or separate sensitivities; the completed marches use the adopted values in its middle column. Section 4 gives the experiment-specific bath histories and initial loadings.

Table 1: Adopted property and transport inputs for the two particle experiments.

Input	Adopted value	Basis and interpretation
Particle radius R_p , mm	0.885 (Faner); 1.50 (DT)	Source mean sphere and coarse end of source size range, respectively (Faner et al., 2019)
Porosity ε_g	0.141	Soybean-meal measurement (Faner et al., 2019); fixed in both baths
Dry meal density $\rho_{dm,p}$, kg/m ³	1.16×10^3	$(1 - \varepsilon_g)\rho_s$; skeletal density $\rho_s = 1.35 \times 10^3$ kg/m ³ (Faner et al., 2019)
Reference critical loading, kg/kg dry	0.20	Source capacity anchor (Faner et al., 2019); event loading uses live liquid density
Critical-volume factor κ_v	2.675	Derived from the 0.20 anchor and reference liquid density 615 kg/m ³
Reference oil fraction $w_{o,ref}$	0.0195	Anchor of the total-meal retention law; DT oil fraction is 0.0137
GAB X_m , kg/kg dry	5.183×10^{-3}	Published soybean sorption fit (Cardarelli & Crapiste, 1996)
GAB C_0 , K_0	3.117×10^{-3} ; 9.172×10^{-2}	Same fit; neither coefficient is adjusted here
GAB $\Delta H_C/R$, $\Delta H_K/R$, K	2262; 729.6	Published temperature dependence (Cardarelli & Crapiste, 1996)
Water isotherm A_1 , kg/kg dry; A_2	0.880; 12.184	Modified Luikov fit (Luz et al., 2006; Pixton & Warburton, 1975)
Water capacity W_{cap} , kg/kg dry	0.236	Upper source-domain anchor; positive-water continuation below 0.0230 is declared
Meal and oil heat capacities, J/(kg K)	2.32×10^3 ; 2.00×10^3	Published meal prior (Coletto et al., 2022); oil value is an engineering estimate
Wet/dry conductivity, W/(m K)	0.24	Porous-mixture prior (Coletto et al., 2022); 0.29 is a separate solid-property sensitivity
Binary pore D_b , m ² /s	6.0×10^{-10}	Assumed 1.5 times the scalar source value (Cardarelli et al., 2002)
Retained-water mobility $L_{w,ret}^{eff}$, mol/(m s)	8.47×10^{-7}	Derived coupled-response value at the pelleted-feed scale (Touffet et al., 2026); consistent with oilseed-meal diffusivities (Cardarelli, 1998; Mukherjee et al., 2019)
Liquid-water mobility $\mathcal{L}_\ell$, mol/(m s)	8.47×10^{-4}	1000 times $L_{w,ret}^{eff}$: water-filled-pore limit of the liquid transport range (Künzel, 1995); outlet on its plateau (factors 100–10 000 marched)
Liquid contact K_ℓ , mol/(m ² s)	0.32	Derived from the 90-s saturation of 1.4-mm solvent-extracted meal at 80 °C (Hemmingsen et al., 2008); construction: Sec. S12
Cross-virial interaction k_{wh}	0.494	Central correlation value; interval 0.477–0.511 (Plyasunov & Shock, 2003)
Pressure, kPa	101.3	Atmospheric reference; 101–170 is the wider process envelope (Kemper, 2013)
Faner film exposure multiplier	0.146	Constant-rate data (Faner et al., 2019), Fig. 2; calculation: Sec. S7.2

3 Numerical implementation and verification

This section explains how the particle equations are integrated in time and how accepted states are checked for admissibility and conservation. The numerical calculation advances component inventories and energy under the prescribed gas state, heat supply and exposure. Surface composition, particle temperature and front motion are solved responses. Each phase regime uses the corresponding equations of Section 2; the numerical scheme transfers the conserved quantities across regime changes.

3.1 Spatial discretization and coupled time integration

Regional balances are integrated in spherical finite volumes. In regime A, the present experiments represent the particle as a uniform thermal body, coupled to the liquid surface through a one-cell conductive resistance. This reduces the coupled startup problem to one body temperature, while retaining a separate surface temperature and conductive heat transfer. It is a coarse approximation, not a spatially resolved preheating calculation; its effect on the initial temperature profile should be assessed before detailed startup heating is interpreted. Regimes B and C resolve the radial particle. The receding interface cuts a fixed radial grid and is represented by the wet volume fraction $z = (s/R_p)^3$. Its swept-volume rate is

$$Q_\Gamma = V_{R_p} \frac{z_{\text{new}} - z_{\text{old}}}{\Delta t}. \quad (28)$$

The same rate enters both regional balances and the interfacial jumps. Backward-Euler integration solves temperatures, compositions, external liquid inventories and front position together. Component enthalpy transport and conductive heat use the same face states as the storage calculation. Detailed face reconstruction and nonlinear controls are given in Sec. S2.

3.2 Phase changes and moving-interface events

When external free hexane is depleted, the dry shell can appear from zero thickness. The first finite-volume interval satisfies the component and energy jumps without assigning an arbitrary initial shell. Arrival at and departure from radial grid faces are solved as geometric events. Near the centre, small positive-core intervals precede a conservative projection to the zero-core particle. That projection includes the external exchanges over its estimated duration; the conditional remaining-time allowance is below 2 ms for DT and 20 ms for Faner. These bounds concern the terminal event, rather than the error of the complete trajectory. External water appearance and depletion likewise switch the surface equations while preserving water and energy. The liquid-water imbibition pathway operates only in a dry shell connected to external liquid. The DT trajectory retains positive external water throughout regime B, so this experiment does not exercise the moving front without a water film. Regime C includes liquid-present, contact and liquid-absent surface states.

3.3 Physical admissibility and conservation checks

At each time step, the nonlinear solution must satisfy non-negative inventories, pressure-feasible phase equilibrium, the receding-front geometry and the retained-water capacity. Capacity is expressed as an active set,

$$0 \leq W_{\text{cap}} - W \perp \lambda \geq 0, \quad (29)$$

with λ its dimensionless dual variable. Local component, energy and combined inventory balances are checked at a normalized tolerance of 10^{-10} . Independent accumulation of the external exchanges provides a separate whole-history check. The four-cell Faner reconstruction uses a reporting allowance

of 4×10^{-10} , corresponding to at most 10^{-11} kg water/kg dry meal; its step acceptance tolerance remains 10^{-10} .

For regular Faner steps, the local nonlinear root criterion can be 10^{-8} while the component and energy acceptance checks retain 10^{-10} . Near wet volume fractions below 10^{-7} , the root criterion can be 2×10^{-8} and the scaled conditioning ceiling 10^{18} . These numerical controls do not change the component or energy equations, their conservation tolerance, or phase admissibility. The Supplementary Material reports the root residual, the conserved-inventory error and the terminal-time estimate separately. Halving the stationary-particle timestep and comparing two and four radial cells quantify particular sensitivities, not whole-history mesh convergence (Section 5).

3.4 Verification of balances and numerical operators

This subsection checks that the discrete operators converge at their expected orders and that the discrete balances conserve water, hexane and energy. Manufactured solutions give spatial orders 1.95–2.00 and temporal orders 0.97–1.00 for the fixed-domain components; the moving-front time operator gives 0.98–0.99. These are tests of individual operators, not measured orders for an entire drying trajectory. Every step of the short refinement ladder satisfies its component and energy balances at or below 10^{-10} in physically normalized units (at most 7.8×10^{-13} on that ladder). The same swept volume in all balances enforces the discrete geometric conservation identity exactly. Shifting the arbitrary enthalpy zero of hexane does not change the solution: the observed datum-invariance difference is 3.42×10^{-10} under its separately stated environment allowance. The discrete identity (16) is the corresponding test for water. It is the check we recommend for any moving-boundary model that claims to count latent and binding heat once (Section 2.4). The full operator-by-operator table, an illustrative numerical sensitivity case that is not a valid limiting case at its initial state, and the unresolved tests are in Secs. S2 and S2.

The discrete caloric-datum test (16) also exposed a consequential interfacial error in an earlier implementation, now corrected. There, the material swept by the front was removed at the equilibrium interface composition but debited from the liquid-hexane core at its bulk composition. Both component and energy must use the same consumed material, as required by (12)–(13); the corrected comparisons use the wet-side material state. These numerical checks establish conservation of the stated equations, not the identity of their unmeasured transport coefficients or predictive accuracy in a plant. The front’s trajectory refinement, including a nonconverged flux-integral result, is reported in Sec. S2.

4 Results

This section applies the particle model to two prescribed baths. The Faner experiment compares solvent loading and temperature with source readings; the DT experiment follows the coupled solvent, moisture and thermal response of a particle from feed to exit. Both use the same particle conservation laws, phase equilibrium, retention closures and reference transport coefficients; their gas histories, geometry and initial states differ. Liquid-water imbibition operates when an external water film contacts the dry shell.

4.1 Falling-rate drying in Faner’s superheated-hexane experiment

Experimental basis and declared particle. The soybean run of Faner et al. (2019) supplies pure n-hexane vapour at 120°C and atmospheric pressure. The article explicitly reports preheating inside the apparatus and no visible warming-up stage. It does not specify the exact initial temperature.

Its product-temperature trace initially lies near the reported 68.7°C boiling plateau. We therefore use a uniform initial body at the calculated pure-hexane boiling temperature, 68.8°C, as a declared approximation to that plateau, rather than a measured preheat endpoint. Initial loadings are $X_h = 0.475$ and $X_w = 0.0250$ kg/kg dry meal, with oil mass fraction 0.0195. Initial moisture is unreported; the positive water loading is a study input that keeps water active in the shared model, rather than a measured property of Faner’s charge. The flakes were desolventized in nitrogen, crumbled and stored at ambient conditions before being re-embedded in hexane. Their initial water therefore cannot be inferred directly from fresh extractor meal moisture. Water loss was not reported separately, so interpreting the gravimetric mass loss as hexane removal introduces an additional observable uncertainty.

The source’s image analysis gives a mean equivalent-sphere diameter of 1.77 mm, mean sphericity 0.74, an overall diameter range 0.5–4 mm and 95.1% of soybean particles between 1 and 3 mm. These summaries do not specify a mass-weighted size histogram. The complete trajectory uses the reported mean diameter directly, giving a spherical radius of 0.885 mm. This choice anchors the particle size to the experiment without adjusting it to the drying curve. It preserves equivalent volume rather than flake shape: a sphere has less surface per volume than a flake of the reported sphericity. The alternative surface-to-volume radius, $0.74 \times 1.77/2 = 0.655$ mm, preserves that area ratio but does not reproduce the flake’s internal transport distances. Neither mapping reconstructs the unreported mass-weighted size distribution. A truncated lognormal construction consistent with the source summaries is documented in Sec. S5; its previous reduced trajectories do not supply the current particle results.

Heat supply and shared model. The film exposure multiplier, 0.146, scales both gas-side heat and mass transfer coefficients while leaving the spherical area unchanged. It was identified from the heat duty implied by Faner’s early constant-rate solvent loss, using an earlier particle-size ensemble, and was not reidentified for the present sphere. The digitized source readings and the calculation are given in Sec. S7.2. The unscaled coefficients describe an industrial reference gas flow, whereas Faner’s apparatus uses a thin sample layer, approximately three to five particle diameters deep (Faner et al., 2019). The multiplier therefore specifies an effective experimental boundary; it is not a measured exposed-area fraction. Scaling mass transfer by the same factor is an additional boundary assumption, since the early heat duty does not identify it independently. The resulting heat coefficient is 15.1 W/(m² K), compared with 104 W/(m² K) before scaling. Thus the early constant-rate agreement involves boundary calibration; the falling-rate and temperature comparisons are conditional on that boundary, with the particle coefficients held fixed. Faner also inferred heat transfer coefficients from experimental drying data (Faner et al., 2019).

For the four separate 2008 thesis runs, Whitaker’s packed-bed correlation predicts 0.82–1.06 times the measured constant-rate duty on the wet-charge mass convention, whereas reading the charge as dry meal gives 0.53–0.61. Two inferred coefficients lie 3.7% and 4.2% below the correlation. These checks concern external supply rather than an independent identification of intraparticle transport (Sec. S3). An otherwise unchanged calculation with multiplier one exhausts external free hexane at 6.8 s, compared with 46.8 s in the reported calculation. This early-stage sensitivity demonstrates the importance of the gas-side boundary; it does not establish that the industrial reference coefficients represent Faner’s apparatus (Sec. S7.2). For reference, Faner’s geometric comparator is

$$s = R_p \left(\frac{X_h - X_e}{X_c - X_e} \right)^{1/3}, \quad (30)$$

where X_e is that source’s equilibrium loading. The present calculation instead solves the component and energy jumps; Eq. (30) is not an additional front constraint.

Both experiments use binary pore diffusivity $6.0 \times 10^{-10} \text{ m}^2/\text{s}$, retained-water mobility $8.47 \times 10^{-7} \text{ mol}/(\text{m s})$ and conductivity $0.24 \text{ W}/(\text{m K})$. These coefficients define the common engineering case; they are not independently identified by the two experiments. Holding them fixed makes the comparison a test of the same transport description in different baths. The positive-water Luikov continuation described in Section 2.2 is used throughout. In regime A, external free hexane is assumed to isolate retained water from the pure-hexane bath: water exchange is zero while its mass and energy remain stored. This surface-access approximation is not a measured barrier property. In regimes B and C, water evolves through the pore-gas and retained-water equations. Liquid-water imbibition is inactive here because there is no external water film.

Figure 2 compares the four-cell calculation with Faner’s readings on the reported experimental clock. Loading includes all particle hexane per unit dry meal; temperature is the particle volume mean, while the thermocouple measures a local holder temperature. Markers show the measurements with their digitization half-widths. The dotted vapour line is the prescribed nominal temperature; the dashed line is the measured vapour trace, shown for comparison. Calculated water and liquid-hexane core volume are supporting results in Fig. S1.

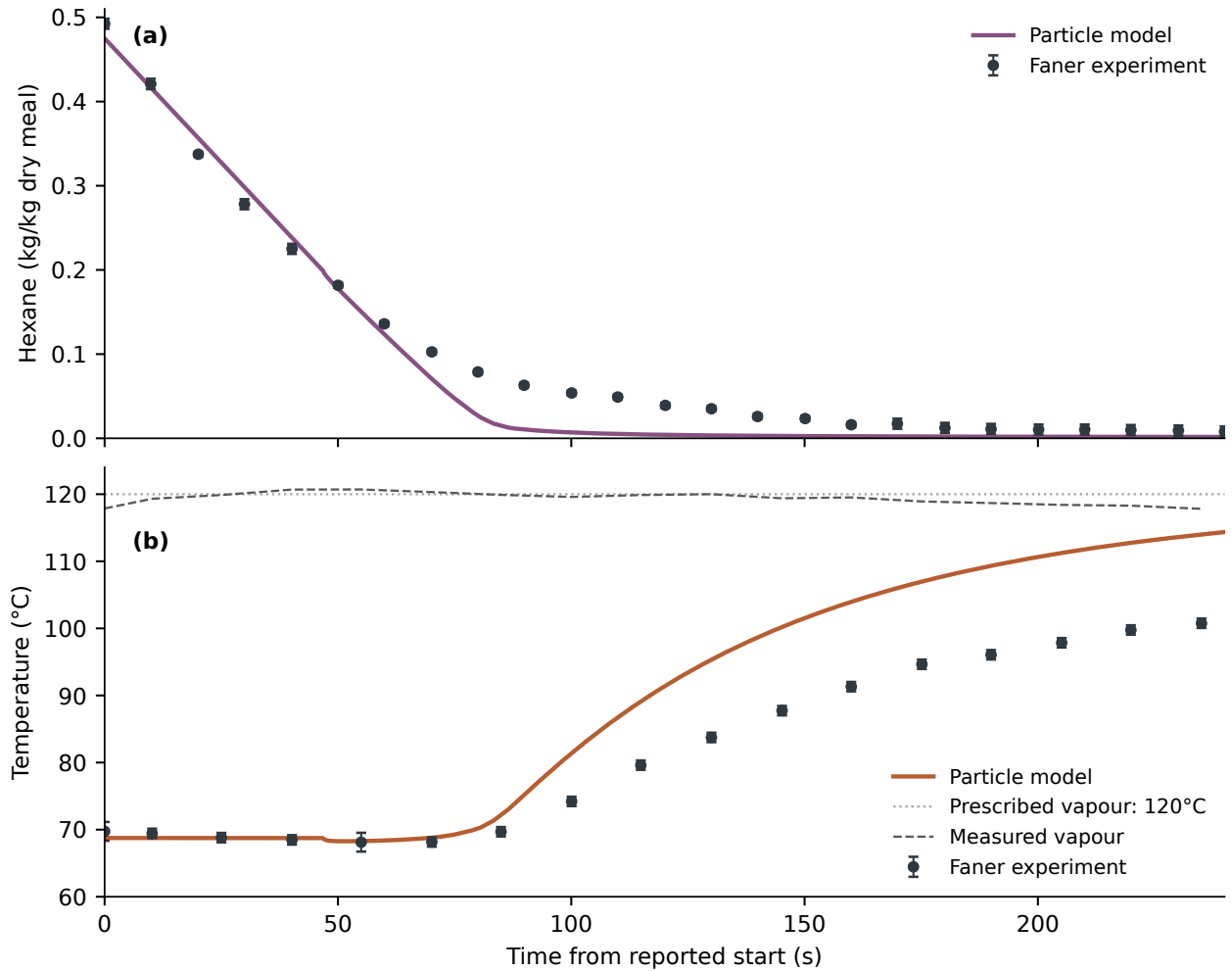


Figure 2: Faner’s pure-hexane experiment: (a) hexane loading and (b) particle temperature. Lines show calculations; markers show measurements.

Solvent removal and temperature. External free hexane disappears at 46.8 s; the core vanishes at 88.1 s. Loading crosses 0.20 kg/kg at 46.6 s and reaches 0.10 kg/kg a further 17.7 s later, versus 45.9 s and 25.4 s in the digitized source. The calculated falling interval is about 30% shorter. At 240 s, hexane is 0.00161 kg/kg dry, water is 0.0144 kg/kg dry and mean temperature is 114.4 °C. This discrepancy cannot be assigned to one mechanism: the particle sizes are known only as summaries, the initial moisture is unknown, and the local exposure and the thermocouple sampling are uncertain. Table 2 summarizes the comparison; loading crossings are interpolated between successive calculated states. Its temperature comparison retains the distinction between local holder and particle mean.

Table 2: Faner comparison on the reported experimental clock.

Observable	Faner experiment	Particle model
Time to $X_h = 0.20$, s	45.9	46.6
Interval $0.20 \rightarrow 0.10$, s	25.4	17.7
Liquid-core disappearance, s	Not measured	88.1
Water at 240 s, kg/kg dry	Not reported	0.0144
Temperature at 235.1 s, °C	100.8	114.0

Using the reported mean diameter instead of the smaller sphere inherited from earlier calculations delays removal and warming; the external area per dry mass decreases and the internal transport distances increase. This improves the loading crossing while leaving a falling-rate discrepancy. At matched four-cell resolution, the 80–100 °C temperature milestones occur about 12–17 s later than with the inherited smaller sphere. In a DT heat balance, that timing determines how long the heat supply goes mainly to evaporation before the meal warms, affecting residence requirements and product quality. The source-mean two-cell march loses its core at 90.1 s; the four-cell value is 88.1 s. Their positive-core step sequences also differ, so this comparison is a resolution sensitivity rather than a continuum limit.

The measured vapour varies from 117.8 to 120.7 °C, rather than remaining exactly at its nominal value. A separate two-cell sensitivity applies that trace after core depletion and gives 113.0 °C at 240 s, compared with 114.3 °C under constant vapour temperature. The small improvement supports treating heat-supply variation as an experimental uncertainty; it does not close the full temperature discrepancy. Before core depletion, this sensitivity still uses the nominal bath (Sec. S5.2). Startup calculations at higher initial moisture, 0.050 and 0.100 kg/kg, are not yet resolved (Sec. S5).

4.2 Particle destiny in the prescribed DT bath

Bath and initial state. The DT experiment uses an effective spherical radius $R_p = 1.5$ mm and follows the particle from feed to the prescribed exit at 1766.5 s (29.4 min). This radius is the coarse end of the previous size screening. It is consistent with the upper end of the 0.5–1.5-mm equivalent-radius range containing 95.1% of Faner’s soybean particles (Faner et al., 2019), but is not a measured DT mean. The longer heat path and smaller surface area per unit mass make this a useful coarse-particle case when examining residence needed for solvent removal. It does not represent a mass-weighted meal size distribution.

Initial temperature is 49.0 °C, hexane and water loadings are 0.388 and 0.124 kg/kg dry meal, and oil mass fraction is 0.0137. The feed is cooler than the 57–60 °C examples reported by Witte (1995) and Kemper (2013), so sensible heating remains part of this experiment. The selected temperature is a scenario input, rather than a measurement for the simulated charge. The pre-desolventizing (PD) bath is at 72.0 °C with hexane mole fraction 0.664, exposure fraction 0.00408 and indirect heat supply 85 W/kg dry until 490.2 s. Steam contact then leads to a toasting bath ending at 105.0 °C. The prescribed gas history is detailed in Sec. S11.

The sequence represents indirect pre-desolventizing followed by live-steam contact, as described by Kemper (2013). Indirect heating removes solvent before steam contact and reduces the solvent load on the steam-heated trays. The selected exposure fraction approximates the small exposed area of a bed compared with the summed areas of isolated spheres. This exposure fraction, the imposed duty and the bath ladder together define the experiment; they do not identify a measured industrial operating point. Liquid-water imbibition uses the closure of Section 2.6, with liquid contact conductance $0.32 \text{ mol}/(\text{m}^2 \text{ s})$, derived from the uptake time of solvent-extracted meal in liquid water, and total dry-shell retained-water mobility 1000 times the reference retained-water mobility, the water-filled-pore limit. Both inputs were set before and independently of the DT outlet.

Liquid depletion, condensation and imbibition. At PD exit, about 0.169 kg external free hexane/kg dry remains. It disappears approximately 2.1 s later while the external water film accumulates. The initial two-liquid surface equilibrium is near 61.3°C ; it differs from pure hexane’s boiling point. The dry shell forms at 492.3 s and core extinction occurs at 503.6 s, approximately 11.3 s after external free-hexane depletion. Retained hexane and pore gas then continue to evolve in regime C. Both the rapid surface evaporation and the core recession follow from the coupled balances.

At core extinction, the particle contains 0.136 kg water/kg dry and the external water film contains a further 0.086 kg/kg dry. The film remains until 624.5 s, about two minutes after dry-shell formation. Liquid-water imbibition and evaporation back to the bath compete for this inventory; condensation does not imply that all incoming water remains in the particle. The external water film supplies mass and energy for several minutes and persists into regime C.

Figure 3 presents wet-basis hexane and moisture alongside the volume-mean particle temperature. Dashed curves use the right axes for the prescribed bath mole fractions and temperature. The lower timeline identifies pre-desolventizing and main trays; dotted vertical lines mark A–B and B–C. The insets resolve the brief core recession and the external water film. Green markers at 30 min show published outlet ranges: 16–22% moisture from Witte and 100–500 ppm solvent and $105\text{--}110^\circ\text{C}$ temperature from Kemper (Kemper, 2013; Witte, 1995). They provide industrial context rather than paired measurements of this prescribed bath. The calculated history ends at 29.4 min.

Exit observables and numerical sensitivity. Particle water at exit is 0.223 kg/kg dry meal, equivalent to 18.2% wet-basis moisture. Hexane is 135 mg/kg dry, or 111 ppm on a wet basis, and volume-mean particle temperature is 105.0°C . The external water film is depleted. The particle-only basis conversion uses $1 + X_w + X_h$; water in the external water film is reported separately while the film is present. Table 3 compares the result with the published outlet ranges. Calculated ppm uses wet meal; the source solvent range does not specify its analytical mass basis. The calculation uses two radial cells and a 5-s maximum regime-C timestep.

Table 3: DT exit results and published outlet ranges.

Observable	Particle model	Published range
Hexane, mg/kg dry	135	Mass basis not specified
Hexane, ppm wet	111	100–500
Particle moisture, % wet	18.2	16–22
Volume-mean temperature, $^\circ\text{C}$	105.0	105–110
External water film, kg/kg dry	0	Not measured

The calculated outlet lies within the published ranges. This agreement cannot identify the imbibition coefficients or validate the prescribed bath. In the reference case at $K_\ell = 0.10 \text{ mol}/(\text{m}^2$

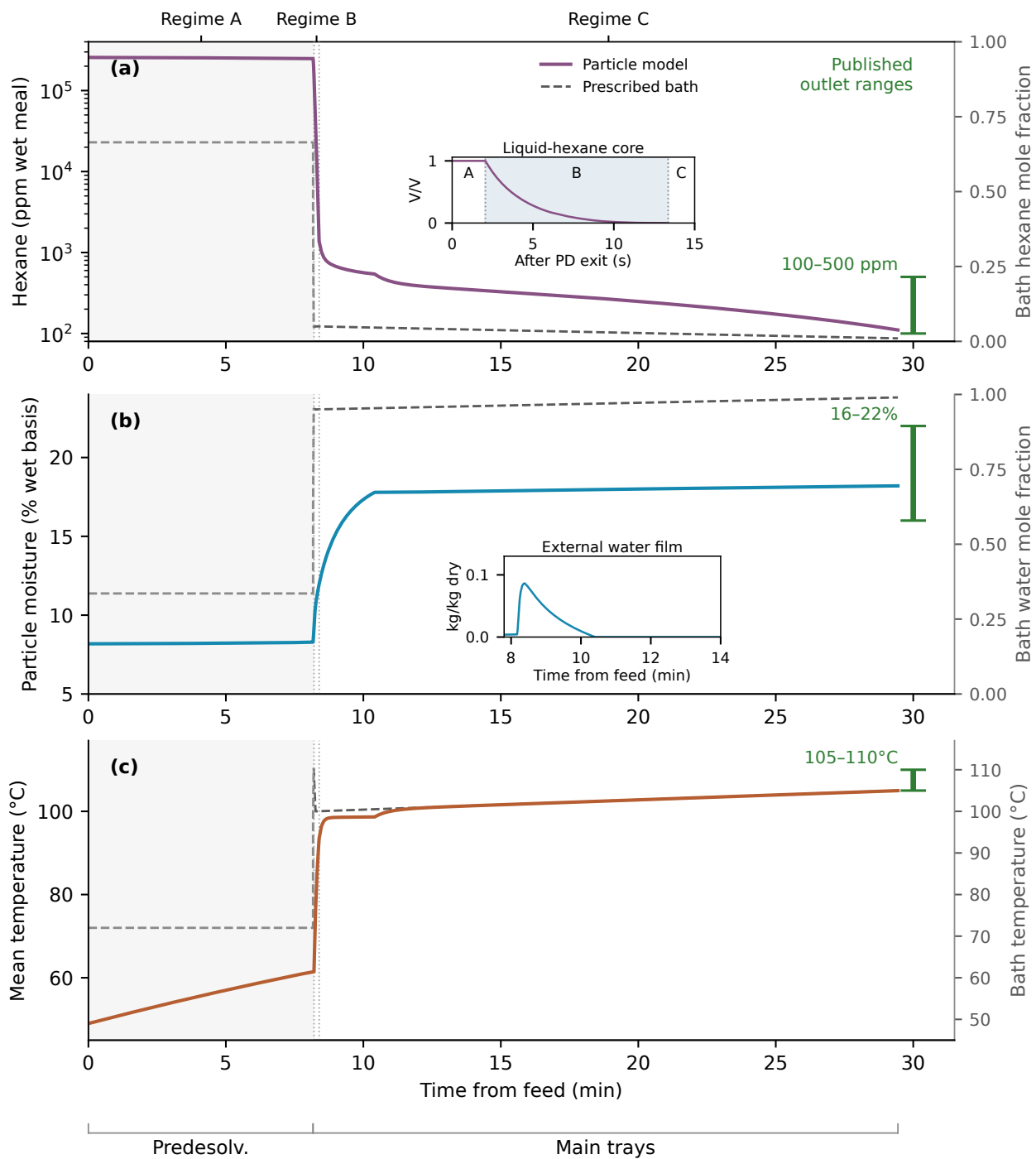


Figure 3: DT particle destiny: (a) hexane, (b) moisture and (c) temperature. Solid lines show the particle response; dashed lines show the bath. Green markers indicate published outlet ranges.

s), halving the regime-C maximum timestep from 10 to 5 s changes moisture by 0.022 percentage points, hexane by 0.52%, temperature by 0.0016 K and water-film depletion time by 1.8 s. The regime-B history is unchanged in this check; full-history mesh convergence is not yet established (Section 5). An independent reconstruction of every accepted advance from dry-shell formation to exit gives maximum normalized water, hexane and energy defects of 4.24×10^{-11} , 8.61×10^{-13} and 6.16×10^{-12} , respectively. The earlier feed audit is retained separately. The coupled result closes the stated balances; the material transport rates remain to be identified independently.

5 Discussion

The two prescribed-bath experiments show how a common particle formulation connects external-liquid evaporation, internal-core recession and residual-component transport. Their signed component and energy exchanges can supply a future tray calculation. Prescribing the bath isolates this particle response, while leaving countercurrent vapour, wall, residence and discharge balances to the equipment model. We first interpret the Faner comparison, then its meaning for DT practice and what is needed next.

The calculated Faner particle still dries faster than the measured charge. Geometry is particularly consequential: image-analysis particle counts cannot be used directly as dry-mass weights, and a volume-equivalent sphere has a different surface area from an irregular flake of equal volume. For a number distribution $p_n(d)$ and common density, the corresponding mass distribution is proportional to $d^3 p_n(d)$. A lognormal family constrained by the reported mean, range and 1–3-mm fraction provides a reproducible sensitivity construction, but does not recover the unreported histogram. Its larger-particle mass fraction can prolong drying even when small particles heat earlier. The direction and size of the resulting temperature change must come from class-resolved integration and the actual thermocouple sampling, rather than from weights selected to obtain a desired displacement of the curves.

Faner reports preheating, so the observed lack of an initial warming stage has an experimental basis. The exact initial temperature and measurement trigger are not specified. A slightly subcooled start would be a different initial-value experiment, with sensible heating included in the balances; it cannot be represented by shifting the calculated temperature alone. The single-particle volume mean also does not determine the temperature of a holder probe. Simultaneous loading, spatial temperature and size-resolved observations would help separate startup, heat exposure and internal transport effects.

For DT practice, the three regimes divide the heat demand. Indirect heating in PD removes external free hexane before live-steam contact; steam condensation then supplies heat for rapid solvent removal, and the remaining retained hexane determines the later approach to the outlet residual. Industrial DT residence and heat supply must serve these successive needs (Kemper, 2013). At the same time, the moist heat treatment must reduce antinutritional activity while preserving useful protein quality (Mustakas et al., 1981; Witte, 1995). The calculated temperature and moisture histories describe that exposure; a meal-quality kinetic model is outside the present scope.

The moisture result also matters for energy design: water retained at DT exit contributes to the load on the downstream dryer, which takes meal towards its storage moisture (Witte, 1995). More condensation can accelerate heat supply and solvent removal, while more retained water can increase later drying duty. A steam-saving claim therefore requires the full balance of steam, indirect heat and drying energy, rather than outlet solvent or moisture alone. These prescribed-bath experiments expose that coupling without quantifying a plant steam saving.

Two independent equilibrium checks give further context for DT practice. At literature-motivated

last-tray compositions, the dry-shell hexane equilibrium floor is 4.5–66.0 ppm without the separate oil-solution sensitivity and 8.1–112.4 ppm with it (Sec. S6). These equilibrium quantities do not set the residence needed to reach them. For retained water the diagnostic relation

$$a_w(X_w)P_{\text{sat}}^w(T^*) = P \quad (31)$$

gives a sorption-elevated boiling range of 100.5–107.6 °C. It combines moisture and temperature observations from different sources; at the moisture of the temperature study, its prediction is 2.1 K above the observation (Sec. S6). Neither this check nor the calculated DT exit moisture is a validation of the coupled transient particle or the industrial tower.

In the DT history, condensate wets accessible hexane-depleted material, and a finite pathway transfers it into the existing retained-water inventory. Experiments by Jovanovich et al. (2003) distinguish liquid-water imbibition from vapour sorption and show that it can include capillarity, matrix hydration and loosely held interstitial water. Their measurements concern protein isolates at 20 °C, not extracted meal in a hot DT. They motivate an effective liquid-connected pathway but do not supply our coefficient, justify its 1000-fold mobility enhancement, or establish that all absorbed liquid is molecularly bound. The present closure keeps one capacity-bounded retained-water inventory and does not resolve those microscopic partitions.

The imbibition closure also omits water-induced pore displacement, a hydraulic saturation front, swelling and extra resistance to hexane escape. In particular, the external water film is assumed permeable to hexane without an additional transport resistance (Section 2.5). Because the film persists for a substantial time in the finite-contact case, this assumption has experimental consequences. Independent transient uptake measurements on the actual extracted meal, with external liquid quantified separately and several particle sizes, are needed to identify the contact and internal resistances. Closer agreement with an outlet moisture range alone would not identify them.

Material and numerical limits set the present scope. The analytic water isotherm is extrapolated below its original 0.023 kg/kg anchor. Hexane retention and binary gas cross properties also have temperature and composition limits, and the transport coefficients are not yet identified from simultaneous water–hexane measurements. Radius and porosity are fixed. The isobaric pore closure omits a total-vapour Darcy/Knudsen resistance and assumes drainage of an immobile connected liquid-hexane core. Numerically, the two-cell DT and four-cell Faner histories provide complete discrete experiments, but operator verification and selected timestep checks do not yet establish whole-history continuum convergence.

6 Conclusions

One conservative particle formulation completes both a 240-s water-bearing particle experiment in pure superheated hexane and a 1766.5-s prescribed DT history. Component and energy balances continue through external-liquid depletion, internal-front recession and core extinction. External condensate is stored separately and exchanges water and energy with the particle and the prescribed gas.

The declared Faner particle falls from 0.20 to 0.10 kg hexane/kg dry meal in 17.7 s, compared with 25.4 s measured. The DT calculation includes liquid-water imbibition and reaches 18.2% wet-basis moisture, 111 ppm wet-basis hexane and a volume-mean particle temperature of 105.0 °C. Its liquid contact conductance is derived from one same-material uptake time and its liquid-water mobility is a declared literature limit; neither is measured on DT meal.

The model’s signed water, hexane and energy fluxes can be coupled to tray balances for DT design; in that coupled form the formulation is the particle-level basis for a DT digital twin that can shorten

process design and support model-based control. The results connect the model’s mechanisms to observable trajectories and show which independent geometry, uptake and temperature measurements are needed before it is used for predictive equipment calculations.

Author contributions (CRediT)

Tomáš Svoboda: Conceptualization; methodology; software; formal analysis; validation; writing – original draft; writing – review and editing. Marcela Sluková: Funding acquisition; resources; supervision; writing – review and editing. Svatopluk Henke: Conceptualization; methodology; writing – original draft. Tomáš Moucha: Investigation; writing – review and editing. All authors have read and approved the submitted manuscript.

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Supplementary material

The submitted Supplementary Material, Secs. S1 to S13, gives the regional and interface derivation, property and benchmark provenance, measured-curve scoring conventions, numerical acceptance and observed convergence results, including the unfavorable cases. An extended technical report retains the longer derivations, former full manuscript passages and research history; no main-text claim relies on that report alone.

Data availability

The governing balances, parameters, comparison conventions and numerical acceptance criteria are stated in the text, the tables and the submitted Supplementary Material. The particle-model source, the scripts that produced the two particle histories and the sensitivity cases, the run records behind every figure and printed number, and the LaTeX sources are released at https://github.com/tomas3svoboda/dt_particle_model (tag v1.0.0-submission, with a SHA-256 manifest of every file); a preservation archive with a persistent identifier is being prepared. Digitized readings of published figures are shared subject to the publishers’ redistribution terms, with reading half-widths and provenance. The steady equipment-model calculation that supplies the prescribed bath of Section 4.2 is not part of the release, but its output is. Numerical material that does not yet exist is listed in Section 5 and Sec. S6; it is not described as available.

Declaration of generative-AI assistance

The authors used AI assistants for language editing, code review and manuscript organization. All physical assumptions, equations, numerical results, claim boundaries and conclusions were formulated, checked and are owned by the authors.

Conflict of interest

Tomáš Svoboda is affiliated with the University of Chemistry and Technology Prague and with Siemens, s.r.o., of which he is an employee; Siemens, s.r.o. had no role in the design of the study, the analysis or the decision to publish. The other authors declare no competing interests.

Nomenclature

Symbols of the equations of this paper, with their SI units; a symbol whose unit depends on the context gives both. Loadings are per kilogram of dry, water- and solvent-free composite meal. Symbols used only in the Supplementary Material are listed there.

Latin symbols

A_1 capacity parameter of the modified Luikov water isotherm (27); kg/kg
 A_2 shape parameter of the modified Luikov water isotherm (27); dimensionless
 a^* pressure-feasible n-hexane activity (1); dimensionless
 a_h n-hexane activity; dimensionless
 a_w water activity; dimensionless
 $\bar{h}_k$ partial enthalpy of component k on the common caloric datum, per mole; $\bar{h}_{h,d}$ that of n-hexane on the dry side; J/mol
 B Darcy permeability of the particle pore space (9); m^2
 B_h binding potential of n-hexane, per kilogram of dry meal (5); J/kg
 C GAB energy constant in (25); dimensionless
 c molar density of the pore gas, in $L_{wh}^{eff} = c y_w y_h D_b$; mol/ m^3
 $c_{h,g}$ n-hexane mass concentration in the pore gas; kg/ m^3
 c_p specific heat capacity at constant pressure; J/(kg K)
 $C_{h,d}^\Gamma, C_{h,w}^b$ n-hexane concentration at the dry-side equilibrium interface trace, and in the wet-side donor state (14); mol/ m^3
 C_k total local concentration of component k over all of its phases; mol/ m^3
 c_w constant shift of the caloric zero of water (16); J/mol
 D_{eff} effective diffusivity of n-hexane in the separate Fickian comparison, Eq. (S2); m^2/s
 D_b binary pore coefficient of the mobility L_{wh}^{eff} ; m^2/s
 d_p mean equivalent particle diameter in the dispersion estimate; m
 e_w specific energy of the liquid-hexane core, per kilogram of dry meal (5); J/kg
 f_{exp} fraction of particle surface exposed to the external gas film in the prescribed DT bath experiment; dimensionless
 h, a, ha external film coefficient per unit particle area; specific external area per dry mass; and their product, the film supply per kilogram of dry meal; W/(m^2 K); m^2/kg ; W/(kg K)
 h_{dm} specific enthalpy of the dry composite meal; J/kg
 ΔP pore-gas overpressure across the dry shell (9); Pa
 $h_{w,l}$ specific enthalpy of liquid water; J/kg
 j_{dm}, j_{Lo} fluxes of the immobile dry meal and residual oil, zero in (3); kg/(m^2 s)
 J_E total radial energy flux; W/ m^2
 J_i binary exchange flux of component i in the pore gas, $J_w = -J_h$ (6); mol/(m^2 s)
 j_k radial flux of component k ; mol/(m^2 s)
 L_ℓ, K_ℓ additional retained-water shell mobility and finite liquid-contact conductance; mol/(m s); mol/(m^2 s)

K GAB constant (25); dimensionless
 k component index, $k \in \{w, h\}$
 k_C regime-C particle-mean hexane relaxation rate, $15D_{eff}/R_p^2$; 1/s
 k_{dry} thermal conductivity of the dry shell; W/(m K)
 k_m packed-bed external gas-film mass-transfer coefficient in the prescribed DT bath experiment; m/s
 L half-thickness of a plane flake; m
 L_{wh}^{eff} binary pore mobility (6); L_{wh}^-, L_{wh}^+ its lower and upper bound; mol/(m s)
 $L_{w,ret}^{eff}$ retained-matrix water mobility (7); $L_{w,ret}^-, L_{w,ret}^+$ its lower and upper bound; mol/(m s)
 M_w molar mass of water; kg/mol
 N_i^g pore-gas molar flux of component i (7); mol/(m^2 s)
 N_t total (Stefan) molar flux of the pore gas; mol/(m^2 s)
 $N_{w,ret}$ retained-matrix water flux (7); mol/(m^2 s)
 N_w^{film} outward-positive water molar flux across the external gas film in the prescribed DT bath experiment, negative for condensation; distinct from $N_{w,ret}$; mol/(m^2 s)
 P total pressure; Pa
 P_{sat}^h, P_{sat}^w vapour pressure of n-hexane and of water; Pa
 q radial conductive heat flux; W/ m^2
 Q_Γ swept-volume rate of the front (28); m^3/s
 R, R_g molar gas constant; J/(mol K)
 r radial coordinate; m
 $r_1(t), r_2(t)$ moving inner and outer radii of a spherical control volume in (10)–(11); m
 r_{pore} pore radius in the capillary-bundle permeability; m
 R_E, R_w discrete energy and water residuals of the interface rows (16); W; mol/s
 R_p particle radius; m
 s radius of the evaporation front; m
 $\dot{s}$ front velocity, negative while the front recedes; m/s
 T temperature; K
 t time; s
 Δt time step; s
 T_{hex}^{bp} boiling temperature of n-hexane in the flashing-zone interpolation of Coletto et al. (2022); K
 T_L flashing-zone solid temperature in that interpolation; K
 T_V vapour temperature in that interpolation; K
 T_g gas temperature; K
 T_Γ interface temperature; K
 T^* boiling point of sorbed water (31); K
 u total local internal energy density; J/ m^3
 u_d^Γ, u_w^b u at the dry-side equilibrium interface trace and in the wet-side donor state; J/ m^3
 $u_{h,l}$ specific internal energy of liquid n-hexane; J/kg
 V_{R_p} particle volume; m^3
 W retained-water loading in (29); kg/kg
 W_{cap} retained-water capacity, the upper end of the measured water evidence; kg/kg
 W_{eq} equilibrium retained n-hexane loading in Eq. (S2); kg/kg

W_{native} retained n-hexane on reference soybean meal (25); kg/kg
 w_o residual-oil mass fraction of the dry meal; dimensionless
 $w_{o,\text{ref}}$ residual-oil mass fraction of the reference meal, the anchor of the total-meal isotherm; dimensionless
 w_h weight on the n-hexane boiling point in the flashing-zone interpolation of Coletto et al. (2022); dimensionless
 W_w retained-water loading of the Luikov isotherm (27); kg/kg
 x reduced activity Ka_h of the GAB isotherm (25); dimensionless
 X_c critical loading, at which mobile liquid stops reaching the outer surface; kg/kg
 X_e equilibrium n-hexane loading at the actual gas activity in the regime-C rate; Faner et al.'s source symbol in Eq. (30); kg/kg
 X_e^* pressure-feasible saturated equilibrium loading at the regime-B/C boundary; kg/kg
 X_f external free-hexane loading; kg/kg
 X_h total n-hexane loading; kg/kg
 $X_{h,w}^a$ apparent mobile-liquid n-hexane loading of the liquid-hexane core, set at activation and carried with the solid; distinct from the total loading X_h and from retained hexane; kg/kg
 X_m GAB monolayer loading (25); kg/kg
 X_w total water loading; kg/kg
 $Y_{w,b}, Y_{w,l}$ bulk and particle-surface gas mass fractions of water in the prescribed DT film closure; dimensionless
 y_i mole fraction of component i in the pore gas; dimensionless
 $y_{w,l}, y_{w,b}$ particle-surface and bulk-gas water mole fractions in the prescribed DT film closure; dimensionless
 z wet-volume fraction $(s/R_p)^3$; $z_{\text{new}}, z_{\text{old}}$ at the new and old time level; dimensionless
Greek symbols
 α region label, w (liquid-hexane core) or d (dry shell)
 β_A Ackermann blowing argument of the film law; dimensionless
 ε_g gas-filled porosity of the dry shell; dimensionless
 θ_a temperature of a material shell at activation; K
 λ multiplier of the capacity condition (29), a contribution to $\mu_{w,\text{ret}}/(RT)$ carrying no mass, volume, enthalpy or internal energy; dimensionless
 μ_g pore-gas viscosity (9); Pa s
 $\mu_w, \mu_h, \mu_{w,\text{ret}}$ chemical potential of water and of n-hexane in the pore gas, and of retained water; J/mol
 ρ gas mass density in the dispersion estimate, $(\rho c_p)_g$; kg/m³
 ρ_f gas mass density used by the external film pair in the prescribed DT bath experiment; kg/m³

$\rho_{\text{dm},p}$ density of the dry composite meal in the particle; kg/m³
 τ pore tortuosity in the capillary-bundle permeability; dimensionless
 ϕ_i fugacity coefficient of component i , omitted by the ideal-gas closure; dimensionless

Subscripts and superscripts

a at activation
 b donor state, the wet-core material the front sweeps (superscript); on $y_{w,b}$, the bulk gas
 c critical
 cap capacity
 d dry shell
 dm dry composite meal
 e, eq equilibrium; eq on d : equivalent
 eff effective
 f external free hexane
 g gas, pore gas
 h n-hexane
 i, k component index
 l liquid
 m monolayer
 native reference soybean meal, at its own oil content
 new, old new and old time level
 o residual oil
 p particle
 ref reference
 ret retained (sorbed) phase
 s surface; on D_s , the solid
 sat saturation
 vap vaporization
 w water; as a region label (on $[\cdot]_w$, $C_{h,w}$, u_w), the liquid-hexane core
 Γ interface; as a superscript, the equilibrium interface trace
 α region, w or d
 $*$ pressure-feasible (on a); boiling point of sorbed water (on T)
 $-, +$ lower and upper bound of an interval

Operators and notation

$[\cdot]_w^d$ jump across the front, dry-side value minus wet-side value
 Δ a difference or a change, as in a time step or an enthalpy of vaporization
 D_s/Dt time derivative following the solid; 1/s
 ∇ radial gradient; 1/m
 ∂ partial derivative
 $\dot{x}$ time derivative of x
 $\perp$ complementarity: both sides non-negative and their product zero

References

Cardarelli, D. A. (1998). *Modelado del proceso de desolventizado de harinas vegetales* (Tesis doctoral). Universidad Nacional del Sur, Bahía Blanca, Argentina.

- Cardarelli, D. A., & Crapiste, G. H. (1996). Hexane sorption in oilseed meals. *Journal of the American Oil Chemists' Society*, 73(12), 1657–1661. doi: 10.1007/BF02517968
- Cardarelli, D. A., Crapiste, G. H., & Mattea, M. A. (2002). Modeling and simulation of an oilseed meal desolventizing process. *Journal of Food Engineering*, 52, 127–133.
- Coletto, M. M., Bandoni, J. A., & Blanco, A. M. (2022). Mathematical modeling and simulation of an industrial desolventizer-toaster. *Journal of Food Engineering*, 318, 110870. doi: 10.1016/j.jfoodeng.2021.110870
- Crank, J. (1956). *The mathematics of diffusion* (First ed.). Oxford; Ely House, London W.1: Oxford University Press / At the Clarendon Press.
- Crank, J. (1984). *Free and moving boundary problems*. Oxford: Clarendon Press.
- Faner, S. A. (2008). *Desolventizado de harinas oleaginosas con vapor sobrecalentado* (Tesis Doctoral en Ingeniería Química). Universidad Nacional del Sur, Bahía Blanca, Argentina.
- Faner, S. A., Perez, E. E., & Crapiste, G. H. (2019). Desolventizing kinetics of oilseed meals with superheated hexane. *Journal of Food Process Engineering*, 42, e12987. doi: 10.1111/jfpe.12987
- Hemmingsen, A. K. T., Stevik, A. M., Claussen, I. C., Lundblad, K. K., Prestløkken, E., Sørensen, M., & Eikevik, T. M. (2008). Water adsorption in feed ingredients for animal pellets at different temperatures, particle size, and ingredient combinations. *Drying Technology*, 26(6), 738–748. doi: 10.1080/07373930802046393
- International Association for the Properties of Water and Steam. (2018, September). *Revised release on the iapws formulation 1995 for the thermodynamic properties of ordinary water substance for general and scientific use* (Tech. Rep. No. IAPWS R6-95(2018)). Prague, Czech Republic: International Association for the Properties of Water and Steam (IAPWS).
- Jovanovich, G., Puppo, M. C., Giner, S. A., & Añón, M. C. (2003). Water uptake by dehydrated soy protein isolates: Comparison of equilibrium vapour sorption and water imbibing methods. *Journal of Food Engineering*, 56, 331–338. doi: 10.1016/S0260-8774(02)00158-9
- Karnofsky, G. (1985, April). Recovering water-immiscible solvents from oilseeds. *Journal of the American Oil Chemists' Society*, 62(4), 693–696. doi: 10.1007/BF03028730
- Kemper, T. G. (2013). Solvent extraction. In W. Hamm, R. J. Hamilton, & G. Calliauw (Eds.), *Edible oil processing* (Second ed., pp. 97–125). John Wiley & Sons, Ltd. doi: 10.1002/9781118535202.ch4
- Krishna, R., & Wesselingh, J. A. (1997). The maxwell-stefan approach to mass transfer. *Chemical Engineering Science*, 52(6), 861–911.
- Künzel, H. M. (1995). *Simultaneous heat and moisture transport in building components: One- and two-dimensional calculation using simple parameters*. Stuttgart: IRB Verlag. (Dissertation, Universität Stuttgart; Fraunhofer Institute of Building Physics)
- Luikov, A. V. (1975). Systems of differential equations of heat and mass transfer in capillary-porous bodies (review). *International Journal of Heat and Mass Transfer*, 18, 1–14.
- Luz, G. R., Sousa, L. H. C. D., Jorge, L. M. M., & Paraíso, P. R. (2006). Estudo das isotermas de equilíbrio do farelo de soja. *Ciência e Tecnologia de Alimentos*, 26(2), 408–413.
- Mukherjee, R., Chakraborty, R., & Dutta, A. (2019). Soaking of soybean meal: evaluation of physicochemical properties and kinetic studies. *Journal of Food Measurement and Characterization*, 13, 390–403. doi: 10.1007/s11694-018-9954-6
- Mustakas, G. C., Moulton, K. J., Baker, E. C., & Kwolek, W. F. (1981, March). Critical processing factors in desolventizing-toasting soybean meal for feed. *Journal of the American Oil Chemists' Society*, 58, 300–305. doi: 10.1007/BF02582363
- Perré, P., Rémond, R., Almeida, G., Augusto, P., & Turner, I. (2023). State-of-the-art in the mechanistic modeling of the drying of solids: A review of 40 years of progress and perspectives. *Drying Technology*, 41(6), 817–842. doi: 10.1080/07373937.2022.2159974

- Pixton, S. W., & Warburton, S. (1975). The moisture content/equilibrium relative humidity relationship of soya meal. *Journal of Stored Products Research*, 11, 249–251.
- Plyasunov, A. V., & Shock, E. L. (2003). Second cross virial coefficients for interactions involving water. Critical data compilation. *Journal of Chemical & Engineering Data*, 48(4), 808–821. doi: 10.1021/jc034046u
- Sipos, E., & Witte, N. H. (1961, March). The desolventizer-toaster process for soybean oil meal. *Journal of the American Oil Chemists' Society*, 38, 11–19.
- Span, R., & Wagner, W. (2003a, January). Equations of state for technical applications. II. results for nonpolar fluids. *International Journal of Thermophysics*, 24(1), 41–109.
- Span, R., & Wagner, W. (2003b, January). Equations of state for technical applications. I. simultaneously optimized functional forms for nonpolar and polar fluids. *International Journal of Thermophysics*, 24(1), 1–39.
- Touffet, M., Délérís, I., & Bokhove, J. (2026). Moisture sorption and diffusion in pelleted animal feed. *Food and Bioproducts Processing*, 156, 474–482. doi: 10.1016/j.fbp.2026.02.001
- U.S. EPA. (1995, November). *Emission factor documentation for AP-42 section 9.11.1: Vegetable oil processing. final report* (Tech. Rep.). Research Triangle Park, NC: U.S. Environmental Protection Agency, Office of Air Quality Planning and Standards, Emission Factor and Inventory Group, with Midwest Research Institute.
- Wagner, W., & Prúß, A. (2002). The IAPWS formulation 1995 for the thermodynamic properties of ordinary water substance for general and scientific use. *Journal of Physical and Chemical Reference Data*, 31, 387–535. doi: 10.1063/1.1461829
- Witte, N. H. (1995). Soybean meal processing and utilization. In D. R. Erickson (Ed.), *Practical handbook of soybean processing and utilization* (pp. 93–116). AOCS Press. doi: 10.1016/B978-0-935315-63-9.50011-5